DOPE.MF sidequest: channel alignment and using ThunderSTORM

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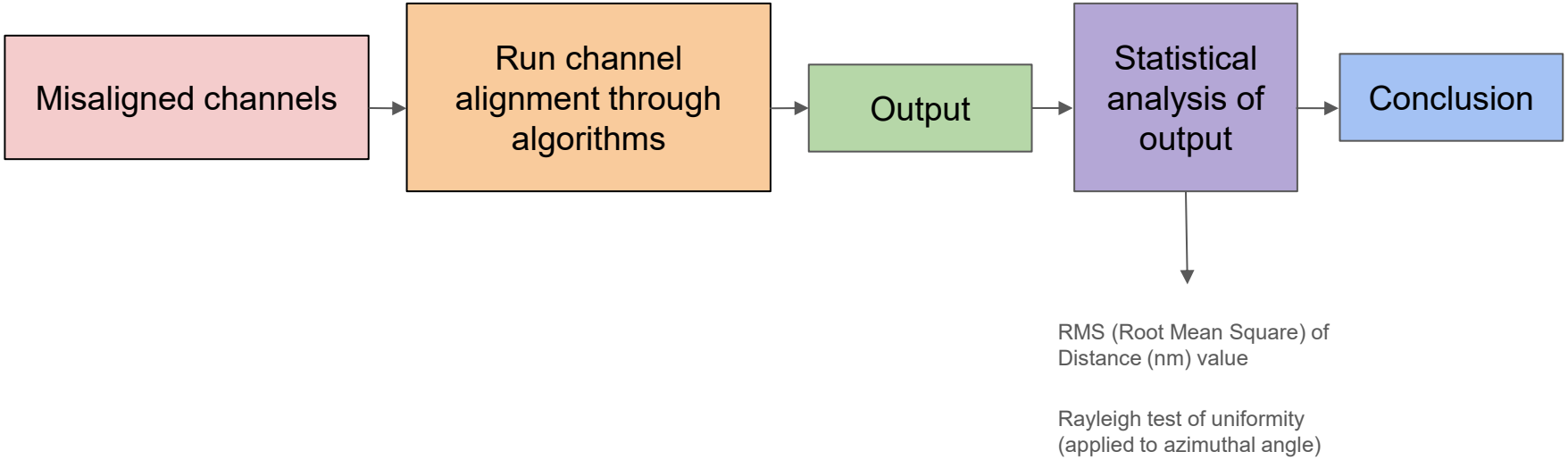
Salaita Lab

Question: Which channel alignment algorithm method provides the best alignment?

We hypothesized that affine was the most accurate alignment technique.



https://pubs.acs.org/doi/full/10.1021/jacs.4c03629

Learning how to use Fiji ImageJ to determine the best form of image registration

Gathering fiducial marker images (ND2 files)

↓

Understanding how to use Fiji ImageJ Descriptor based registration software

↓

Performing thunderstorm analysis on the output files

↓

Finally did the analysis using python to get finalized results for what is the best form of image registration and to quantify the best format

↓

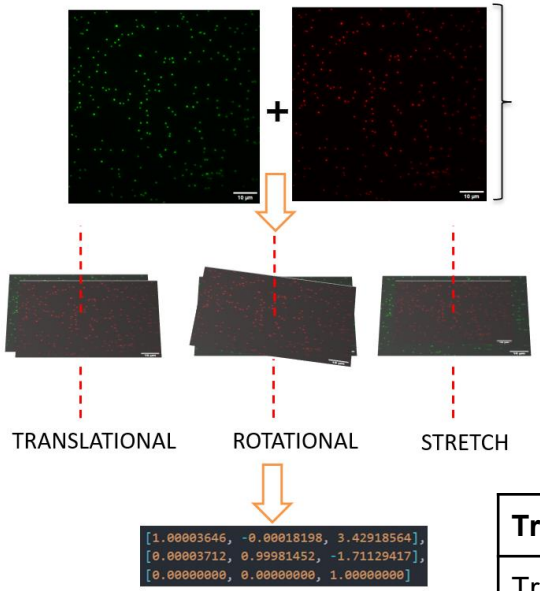
Conclusion: Homography algorithm.

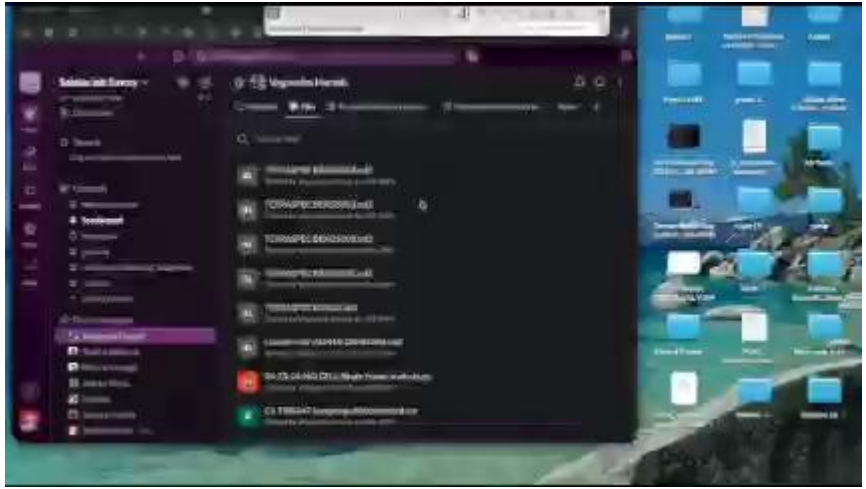
Wrote a handbook and it will be uploaded to GitHub and also made recording to go along with the handbook

1. Image registration techniques
   a. Translation (2d)
   b. Rigid (2d)
   c. Similarity (2d)
   d. Affine (2d)
   e. Homograph (2d)

2. Incorporation of ThunderStorm (https://zitmen.github.io/thunderstorm/) for single-molecule light (SMLM) data analysis

Image registration for alignment : Descriptor-based Registration (software)

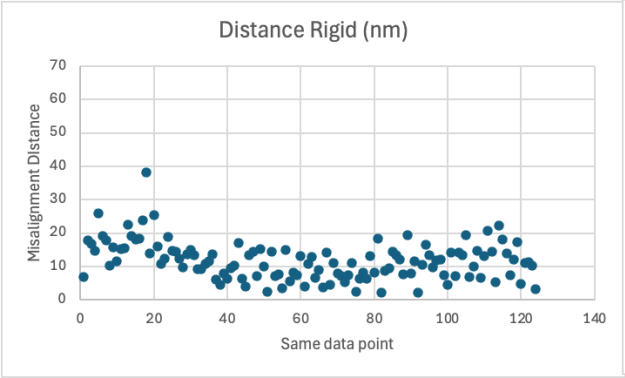


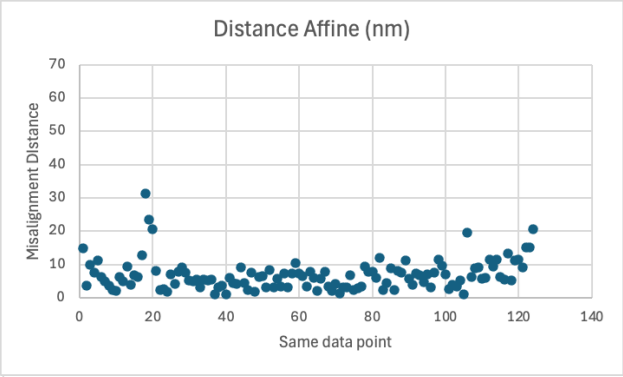


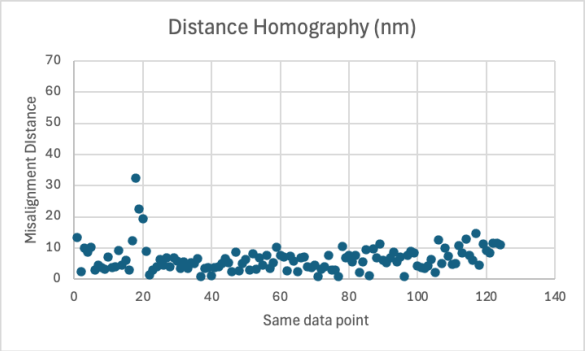
Video: tutorial on how to perform image registration and ThunderSTORM analysis

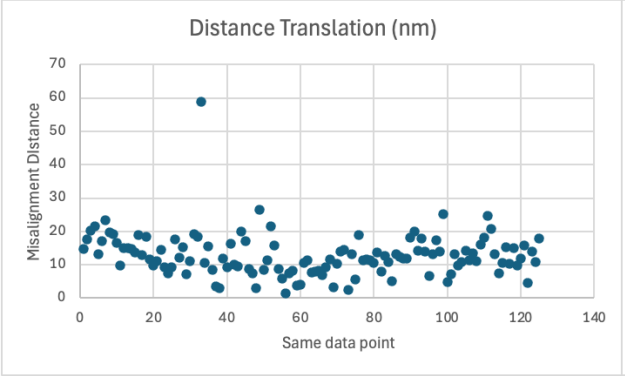
Transformation	Translation	Rotation	Scaling/stretch	Shear(tilt/distortion)	Perspective
Translation	YES	NO	NO	NO	NO
Rigid	YES	YES	NO	NO	NO
Similarity	YES	YES	YES(uniform)	NO	NO
Affine	YES	YES	YES(non-uniform)	YES	NO
Homography	YES	YES	YES	YES	YES

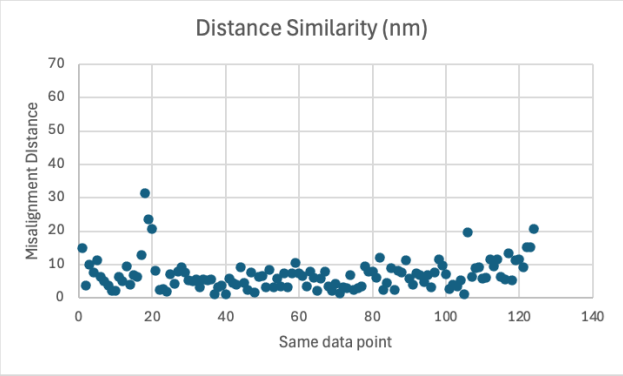
Comparison of alignment algorithms: Distance (x-axis: same data point, y axis is **misalignment** distance between red and green channel in nanometers)



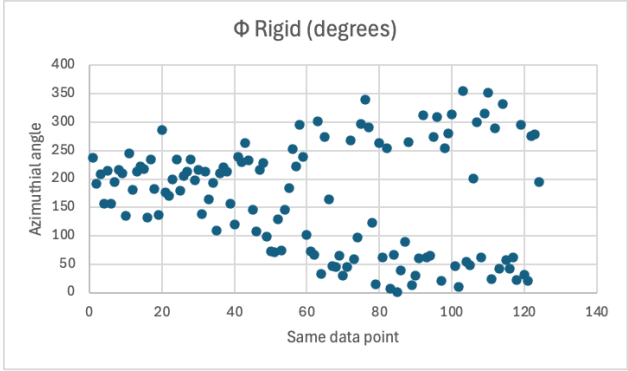


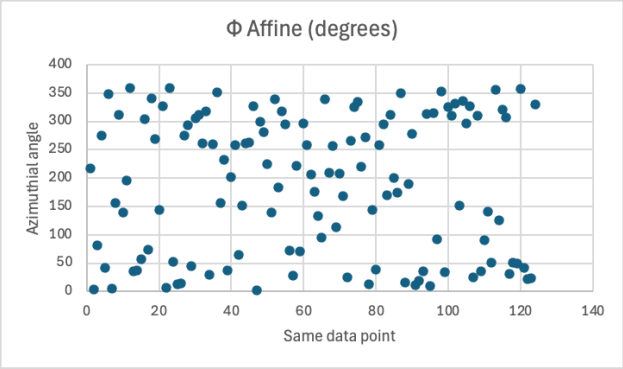


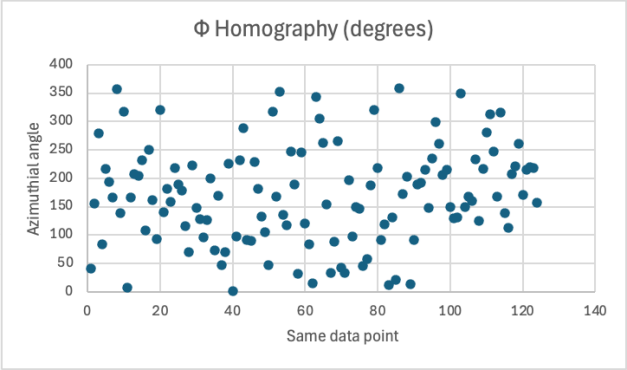


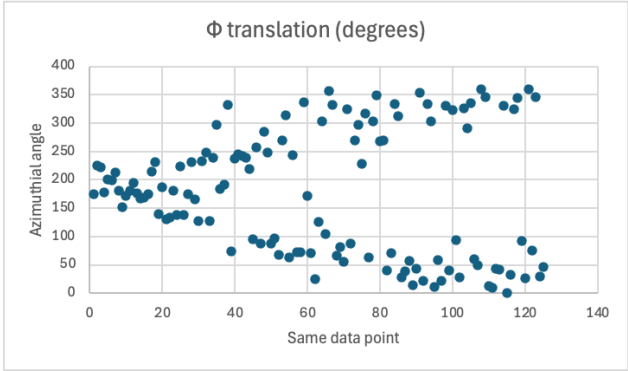


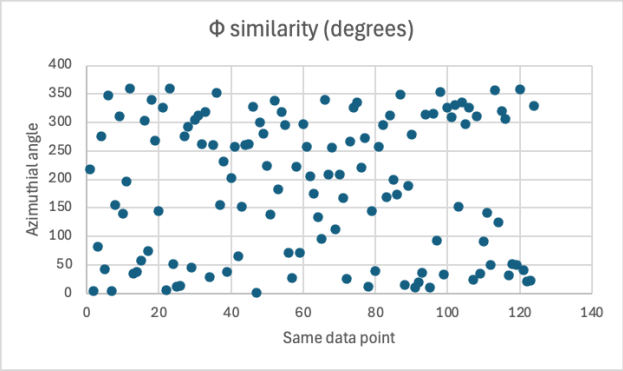
Comparison of alignment algorithms: Azimuthal angle (x-axis: same data point, y axis is change in Azimuthal angle between red and green channel











RMS (Root mean Square) of distance values

Method	RMS(nm)	Median +/- STD (nm)	Final conclusion
Homography	7.76	5.70+/- 4.2	Lowest RMS value
Affine	8.41	6.05+/-4.73	Lower RMS value
Similarity	8.41	6.05+/-4.73	Lower RMS value
Rigid	12.97	11.37+/-5.66	Higher RMS value
Translational	14.40	11.82 +/-6.55	Highest RMS value

Null hypothesis: Residual angles are uniformly distributed with no preferred direction.

Rayleigh Test test of Uniformity for azimuthial angles

Method	Rayleigh Z	p-value	Final conclusion
Homography	0.96	0.38	Error minimized, No directional bias (p>0.05)
Affine	1.07	0.34	Error minimized, No directional bias (p>0.05)
Similarity	1.07	0.34	Error minimized, No directional bias (p>0.05)
Rigid	0.88	0.41	smaller errors, No directional bias (p>0.05)
Translational	10.00	4.5×10^{-5}	Larger errors, directional bias (p<0.05)

Conclusion

	Low RMS residual distance	High RMS residual distance
Non significant Rayleigh Test	Homography !	
Significant Rayleigh Test		

Homograpghy minimizes spacial error even with slight directional bias ($p<0.001$) as per the Rayleigh Test

Handbook and Github page

Channel-alignment-and-ThunderStorm-analysis-DOPEMF Public

This repository outlines a workflow for channel alignment using ThunderSTORM localization data. It includes preprocessing, transformation-based registration, and analysis scripts to quantify misali...

Python 1